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Lehrbuch der Algebra
Working English Translation Batch 10
Absolute Invariants and the Form Problem

Translated into modern mathematical English

Translator's note

This batch continues Weber's invariant-theoretic bridge between linear groups and equation theory. The term "form problem" translates *Formenproblem*.

§42. The absolute invariant and the ground field

Let

$$F_1, F_2, \ldots, F_r$$

be a basis of the invariant system J, as determined above, and let F be any other nonconstant invariant of S. The forms $A_1, A_2, \ldots, A_r$ may then be chosen so that

$$F = A_1F_1 + A_2F_2 + \cdots + A_rF_r. \tag{1}$$

Apply all substitutions of the group S to this identity. By assumption,

$$F, F_1, F_2, \ldots, F_r$$

remain unchanged; the coefficients $A_1, A_2, A_3, \ldots$, however, may be transformed into

$$A_1^{(i)}, A_2^{(i)}, A_3^{(i)}, \ldots.$$

If one adds all identities obtained in this way and divides by the order s of the group S, then one obtains

$$F = \overline{A}_1F_1 + \overline{A}_2F_2 + \cdots + \overline{A}_rF_r, \tag{2}$$

where now $\overline{A}_1, \overline{A}_2, \ldots$ are themselves invariants of the group S. By the defining property of the basis, these averaged coefficients are rationally expressible through $F_1, \ldots, F_r$.

Hence:

Theorem. *Every invariant F can be represented as a rational function of the basis invariants $F_1, \ldots, F_r$.*

Let

$$\vartheta_1, \vartheta_2, \ldots, \vartheta_h$$

be the coefficients of the general substitution of S, and let Ω be the rational field generated by these quantities. Every invariant F is then a function of the variables x and of the quantities

ϑ. If the x's are treated as independent variables and the ϑ's as parameters, the totality of all invariants so obtained forms a field. We shall call it the *field of invariants* of the group S.

The absolute invariants are precisely those functions that remain unchanged under all substitutions of the group and depend only on the orbit of the point under the group action. Thus they furnish the rational functions on the quotient of the space of variables by the group.

§43. The form problem

The problem of determining the values of

$$x_1, x_2, \ldots, x_n$$

which give a prescribed value F_0 to a given invariant F, when the coefficients of the group are assumed known, is called the *form problem* of the group S.

Let $\Omega(t)$ denote a general function of the field Ω which is carried into itself by the substitutions of S. The roots of the equation

$$\Omega(t) = \omega,$$

where ω is any fixed value, are all distinct and are transformed into one another by the substitutions of S.

The solution of the form problem is closely connected with the solution of the equation

$$\Omega(t) = \omega.$$

Indeed, if ϑ is a root of this equation and one sets

$$x_i = \vartheta_i(\vartheta), \tag{3}$$

for suitable functions ϑ_i, then one obtains values of the x_i satisfying all invariant relations.

If the invariant J can be represented as a rational function of ϑ, then this function cannot change when any of the substitutions

$$(\vartheta, \vartheta_1), \quad (\vartheta, \vartheta_2), \quad \ldots$$

is applied. Consequently it is rationally expressible through the coefficients of $\Omega(t)$. Whether this representation is possible depends on the nature of the group.

If S' is a subgroup of S of index j, and no larger subgroup has the same invariance property, then ϑ satisfies an equation of degree j. This equation is to be regarded as a *resolvent* of the form problem. Every function which admits exactly the substitutions of S' satisfies such a resolvent.

Thus the form problem transforms invariant theory into an equation problem: the passage from a group to its subgroups corresponds to the passage from a general equation to its resolvents.